

Introduction

Hi, my name is Kyra Wilson, I use she/her pronouns, and today I'm going to be talking about our project entitled "Contextual Embeddings Can Distinguish Homonymy from Polysemy in a Human-Like Way."

So the first thing I want to do in this presentation is introduce or remind everyone about what homonymy and polysemy actually are. Both of these are types of lexical ambiguity—where a word has the same spelling or sounds, but possibly multiple different meanings. Polysemy is when these meanings are related to each other—so in this instance the word 'paper' can mean a sheet for writing on, or it might mean something you get the news from. These both share some semantic aspect though, perhaps related to the material of the object. Importantly, when these meanings are related, they are often referred to as senses and that is the terminology I will use for the rest of this presentation. Homonymy, on the other hand, has unrelated meanings. 'Bark' for example has two meanings—one that has to do with the noise a dog makes and one relating to the substance on the outside of a tree. Although these meanings are said and spelled identically, they are not related semantically. It's also important to note here that a word can have both unrelated meanings and related senses. If we take the dog meaning of the word 'bark' it also has two related senses—the sense that refers to the action of the dog barking (a verb) and the sense that refers to the sound of the bark (a noun). In this way, senses and meanings can be thought of as having a kind of hierarchical structure—each word will have one or more meanings, and under each of those meanings is the potential for additional related senses.

Now when we think about how language models deal with lexical ambiguity and disambiguating intended meanings, typically the notions of homonymy and polysemy are not relevant. Most current models will use some kind of contextual embedding, where the representation of a word is dependent on the words surrounding it. Whether a word has multiple meanings or senses is irrelevant to the model, and so they are both treated identically.

According to psycholinguistic research, this does not seem to be the case for humans, and the distinction between homonymy and polysemy does seem to be relevant. This has mostly been observed in the lexical decision paradigm, where subjects are asked to read a word and quickly decide if it is real or nonsense. In these studies, experimenters observe that polysemous words with related senses are recognized more quickly than unambiguous words, and homonymous words with unrelated meanings are recognized either as quickly as unambiguous words or slower than them.

Research into this so-called "ambiguity advantage" is ongoing, but it faces severe limitations. Namely, results are often inconsistent between studies because there is no standard criteria for deciding whether words are homonymous or polysemous. Frequently resources like dictionaries or sense inventories like WordNet are used, but these are human-curated and could be biased in some way. Additionally, studies are not consistent in which resources are chosen, so words classified as polysemous in one study may be homonymous in another, or vice versa.

This brings us to the main objectives of this investigation—we are interested in leveraging the success of contextualized embeddings ability to distinguish meanings of words and applying it to the analysis of human behavioral data. Namely, is it possible to use contextual embeddings to approximate a measure of polysemy and homonymy? And if so, can this measure be used to predict human behavior in a way that replicates previous findings about the ambiguity advantage?

Ambiguity Measures

First, we worked on deriving a measure of polysemy. The first major hurdle with using contextual embeddings to quantify ambiguity was that they are, by nature, contextual while human behavioral data often lacks context because words are presented in isolation. We needed to find a way to dissociate contextual embeddings from their context. Broadly, this led us to the approach of sampling a large number of occurrences in their contexts to retrieve a contextual embedding, and then applying a clustering algorithm to the set of individual word embeddings to derive an approximation of the number of senses. Intuitively, under this approach words that are used similarly to each other should be close together in space and words that are used differently should be further apart. This way a clustering algorithm will be able to identify the individual senses of the words.

More specifically, following research from a team at Google Brain, we sampled 1,000 occurrences of a word of interest and the sentences in which they appeared from English Wikipedia.

And following that we fed the sentences to a BERT_{BASE} model to get a context-dependent embedding of the word. At this point, we simply extracted the token embedding of the word from the final layer, and discarded the rest of the sentence embeddings as well as weights from previous layers.

Next we applied the tSNE algorithm to the embeddings to reduce the dimensionality from 758 to 2. We did this for two reasons—first, the clustering algorithm we used performs poorly on extremely high dimensional data. Second, we wanted to be able to judge our measures qualitatively as well as quantitatively, and so it was important to be able to have embeddings which could be easily visualized.

The final step in deriving a number of senses was simply applying the HDBSCAN algorithm to the set of word embeddings. One benefit to this algorithm is that it does not require a priori specification of a number of clusters. The only required hyperparameter is a minimum cluster size, which we set as 1% of the total sample size, so 10 points. An additional benefit of this algorithm is that not all points must be assigned to a particular cluster, some may be identified as noise. This is particularly useful because language is very flexible and dynamic. Individual usages of a word may be particularly idiosyncratic and not a good representation of the word as a whole, so it is useful to be able to discard these points in the estimation of a number senses so that it does not become artificially inflated.

The number of senses a word has (our measure of polysemy) is equivalent to the number of clusters identified. Now we are able to work on the estimation of the number of meanings. If you recall, meanings and senses can be thought of hierarchically, and we wanted to replicate this in our approach to deriving a number of meanings. We operationalized this as finding superstructure amongst the sense clusters we derived. Are there sense clusters which are closer together that might form a shared meaning, while others which are further apart indicate unrelated meanings? We decided to approach this question in the same way we approached the identification of senses—by using a clustering algorithm.

First, to identify any superstructure we wanted to select the most representative points from sense clusters. The HDBSCAN algorithm identifies exemplar points in the clustering process, those which are the most sustained members of the cluster. Doing this allowed us to eliminate any additional noise points included in a cluster and also provided us a different set of points to cluster so we would not simply replicate the senses result.

Finally we applied the same HDBSCAN clustering algorithm to the set of exemplar points. We also increased the minimum cluster size to be equal to the smallest number of exemplar points identified within a single cluster. This way we biased the algorithm to identify relationships between the clusters rather than structure within them.

The number of second-order clusters is what we use as our number of meanings, which is the measure of homophony.

Example Clustering

The results of this procedure can be illustrated using an example like the word 'tent'. First, we can see that the first clustering procedure resulted in three senses being identified. The first cluster, made of square points, includes occurrences of 'tent' as a proper noun. The second, made of circular points, has instances of a literal tent structure. Finally, the third, made of crosses, has more metaphorical usages of 'tent'. These do all seem like distinct senses. Next for the clustering of meanings, we can see exemplar points that are identified are the largest sized points. There are a few in each of the clusters.

When we apply the clustering algorithm to the exemplar points, only one cluster is found indicated by the color purple. This collapses over the structural and metaphorical 'tent' usages, which makes sense because they probably both share some semantics about a gathering space for people. The first sense cluster is excluded from this meaning, probably because the proper noun usages are more idiosyncratic and do not necessarily share semantics with the other two senses.

Experiment and Results

To evaluate the performance of our measures in predicting human behavior, we aimed to replicate a previous experiment that showed the classic ambiguity advantage effects.

The data used in this study were the original 182 words in the study which included 124 ambiguous words and 58 unambiguous words. We retrieved reaction times from the English

Lexicon Project, a database of English words and their mean reaction times in a lexical decision megastudy.

We used a linear regression to predict the reaction time as was done in the original study. In addition to the polysemy and homonymy variables, there were also a number of control variables including log word frequency, length, orthographic neighborhood, and concreteness.

For the control variables, the only that had an effect were the log word frequency and length. We did not replicate the concreteness and neighborhood effects that were found in the original experiment.

The original experiment did not specify how the number of senses were calculated, so we used WordNet to approximate their measure. We did not find an effect of WordNet senses, even though they did report a sense effect in their original experiment. Additionally, this experiment did not calculate the number of meanings, but rather used a binary variable indicating whether words had a single or multiple meanings. In our replication we did find that this variable had an inhibitory effect as was reported initially, meaning that words with more than one meaning were responded to more slowly than words with one meaning.

With our measures derived from BERT, however, we were able to replicate both of the original reported results.

Our number of senses measure had a facilitatory effect, meaning words with more senses were recognized more quickly than words with fewer senses.

We found the opposite effect for the number of meanings, where words with more meanings were recognized more slowly than those with fewer senses. This does replicate the ambiguity advantage as it has been described in the psycholinguistic literature.

We also performed an additional ANOVA analysis to compare the contribution of our measures to the fit of the model. We compared linear models with and without the number of meanings and number of senses, and in both cases the fit of the model was significantly improved by adding the variables derived from BERT.

Conclusion

So as I just mentioned, the first important result of this study is that it is possible to quantify the polysemy and homonymy of a word using only contextualized word embeddings, specifically BERT.

Additionally, these measures are predictive of human behavior in lexical decision tasks, and they replicate the ambiguity advantage effects that have been found previously, where multiple senses ease recognition and multiple meanings make it more difficult.

Both of these results mean researchers interested in the ambiguity advantage will have options to derive predictors outside of the traditional human-curated resources. Additionally, because we

have a clear methodology and criteria for defining senses versus meanings, results between experiments may be more consistent when using our methods. Most research into the ambiguity advantage has been done in English, but this methodology also expands the range of languages that can be investigated. It is not necessary to have individual dictionaries or sense inventories for every language, all that is needed is the ability to train a model for the language of interest.

I would like to thank all of you for listening to our presentation as well as our funders and the many people who have given advice and support for this project. If you'd like to view a PDF of this poster or the code used in our experiment, feel free to click on the link or scan the QR code. If you have any additional questions, please reach out to me!